

Intrinsic invariants of $(3, 4)$ -cuspidal edges

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Introduction: Why $(3,4)$ -cuspidal edges?

- $(3,4)$ -cuspidal edges arise naturally on **parallel surfaces**.
- Despite their natural appearance, their geometry is less understood than that of ordinary cuspidal edges.

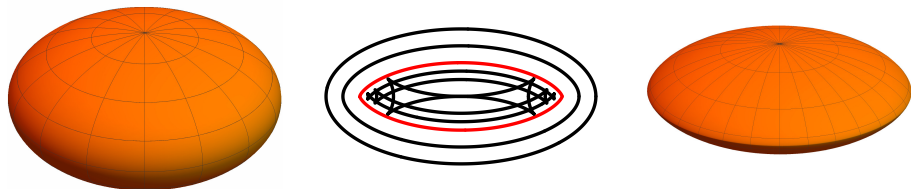


Figure: An ellipsoid (left), a parallel curve with a $(3,4)$ -cuspidal edge (center, red), and a parallel surface of the ellipsoid (right).

Introduction: Difference from cuspidal edges

Invariant and intrinsic invariant

In this talk,

- **Invariant**: its absolute value is preserved under coordinate changes and Euclidean isometries.
- **Intrinsic invariant**: determined only by the **first fundamental form**.

Cuspidal edge

The singular curvature κ_s is an **intrinsic invariant**, and its sign is well-defined.

$\text{sgn } \kappa_s$ $\longleftrightarrow$ local shape



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Cuspidal edge (SUY. 09 8/18) (3,4)-cuspidal edge

The singular curvature κ_s is an **intrinsic invariant**, and its sign is well-defined.

$\text{sgn } \kappa_s \longleftrightarrow \text{local shape}$



The sign of κ_s depends on the orientation of the null vector field:

$$\eta \mapsto -\eta \implies \kappa_s \mapsto -\kappa_s.$$

Hence its sign is ~~generally~~ **not well-defined**.

($\kappa_s = 0$ is well-defined)

κ_s is

For $(3,4)$ -cuspidal edges, the sign of κ_s is **NOT** well-defined.

Question

Can we construct intrinsic invariants that describe the geometry of $(3,4)$ -cuspidal edges?

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Gaussian curvature

When is K bounded?

Shape

Convex or saddle?

Singularity type

Can we characterize $(3, 4)$ and $(3, 5)$?

3-type edges: a class containing (3, 4)-cuspidal edges

Let Σ be a two-dimensional manifold.

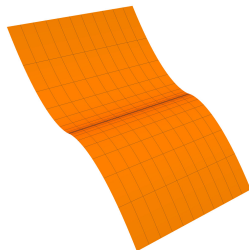
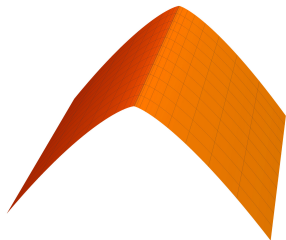
Definition(3-type edges) Martins–Saji–Santos–Teramoto (2024)

A smooth map $f : \Sigma \rightarrow \mathbb{R}^3$ has a **3-type edge** at $p \in \Sigma$ if the map-germ $f : (\Sigma, p) \rightarrow (\mathbb{R}^3, f(p))$ is $\mathcal{A}$ -equivalent to

$$(u, v) \mapsto (u, v^3, v^4 \alpha(u, v))$$

for some smooth function α .

(u, v^3, v^4) : (3, 4)-cuspidal edge, (u, v^3, v^5) : (3, 5)-cuspidal edge.



Degeneracy order of the signed area density

Let $\lambda = \det(f_u, f_v, \underline{\nu})$ be the signed area density function.

New in this work

7D-型ルを定義できる.
(Appendix 1F3) 3-type edge $\Rightarrow$ 7D-型ル

Definition (singular point of degeneracy order n)

Let $n \geq 1$. A rank 1 singular point p has **degeneracy order** n if there exist smooth real-valued functions $\hat{\lambda}$ and α such that

$$\lambda = \alpha \hat{\lambda}^n, \quad d\hat{\lambda}(p) \neq 0, \quad \alpha(p) \neq 0.$$

- $n = 1$: the usual non-degenerate case.
- We focus on $n = 2$.

In the following, “**the first kind**” means that the null direction is transverse to the singular set.

Degeneracy order 2 and 3-type edges

Proposition

Let p be a singular point of the first kind of a frontal $f : \Sigma \rightarrow \mathbb{R}^3$. Then

p has degeneracy order 2 $\iff f$ has a 3-type edge at p .

Thus, 3-type edges can be viewed as
second-order degeneracies of the signed area density.

NACS: a convenient coordinate system for the metric

In this talk, we write

$$\bullet G_{V^j} := \frac{\partial^j G}{\partial V^j}.$$

$$\bullet ds^2 = E du^2 + 2F du dv + G dv^2, \quad \bullet S(f) = \{v = 0\}.$$

Normalize ds^2 into a form that is easy to compute with.

Normalized adapted coordinate system (NACS)

A local coordinate system (u, v) is called a **NACS** if

$$\begin{aligned} E(u, 0) &= 1, & F(u, v) &\equiv 0, \\ G(u, 0) &= G_v(u, 0) = G_{vv}(u, 0) = G_{v^3}(u, 0) = 0, \\ G_{v^4}(u, 0) &= 6. \end{aligned}$$

Existence

A NACS exists near every singular point of the first kind of degeneracy order 2.

Intrinsic quantities β and δ

The higher derivatives of E and G depend on the choice of NACS.

New in this work

Definition(β, δ)

Along $S(f) = \{v = 0\}$, define

$$\beta(u) := E_{v^4} - \frac{G_{v^5}}{20} E_{v^3},$$

$$\delta(u) := E_{v^5} - \frac{G_{v^5}}{12} E_{v^4} + \left(\frac{7G_{v^5}^2}{1200} - \frac{G_{v^6}}{30} \right) E_{v^3}.$$

(All terms on the right are evaluated at $(u, 0)$.)

For another NACS,

$$\hat{\beta} = \beta, \quad \hat{\delta} = \pm\delta.$$

So, β and δ are **intrinsic invariants**.

What do β and δ measure?

Let K be the Gaussian curvature on the regular set.

Expansion of $v^2 K$

In a NACS, ~~by the Theorema Egregium,~~

$$v^2 K = -\frac{1}{3}\beta + \left(-\frac{1}{6}\delta + \frac{G_{v^5}}{360}\beta\right)v + \mathcal{O}(v^2).$$

$\beta \neq 0$

At a fixed point of the singular set,

$$K = -\frac{\beta}{3v^2} + \mathcal{O}(v^{-1}).$$

β controls the **leading divergence**.

$\beta = 0$

If $\delta \neq 0$, then

$$K = -\frac{\delta}{6v} + \mathcal{O}(1).$$

δ controls the **next divergence**.

Main result I: boundedness of Gaussian curvature

(仮定) f : 3-type edge
前提を正確に書く.

Theorem A

Let $I \subset S(f)$ be a sufficiently small interval containing p . Then

$$K \text{ is bounded in a neighborhood of } I \iff \beta \equiv 0, \delta \equiv 0 \text{ on } I.$$

The first two coefficients of $v^2 K$ give a complete criterion for boundedness.

Main result II: intrinsic detection of the local shape

In the following results, generic means $\kappa_\nu(p) \neq 0$.

Proposition B

For a generic $(3,4)$ -cuspidal edge at p ,

- f is saddle-shaped at $p \iff \beta(p) > 0,$
- f is convex at $p \iff \beta(p) < 0.$

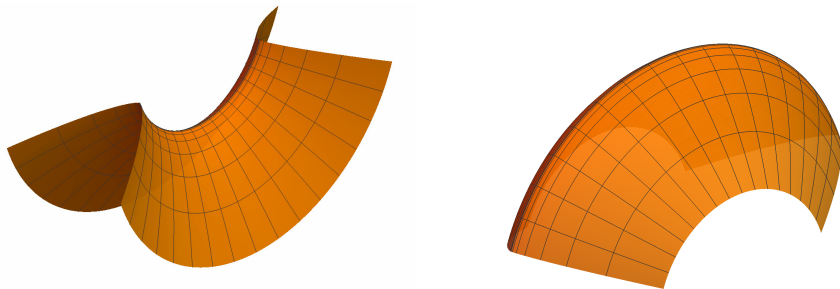


Figure: $\beta > 0$ (left: saddle-shaped) and $\beta < 0$ (right: convex) along the singular set.

Bridge to known singularity criteria

For 3-type edges, Martins–Saji–Santos–Teramoto introduced the cuspidal curvatures

$$\omega_{3,4}, \quad \omega_{3,5}.$$

which measure higher-order cuspidal behavior.

- $\omega_{3,4}(p)$ is an invariant.
- If $\omega_{3,4} \equiv 0$ along the singular set, then $\omega_{3,5}(p)$ is also an invariant.

Key relations

Under the genericity assumption,

$$\beta(p) \neq 0 \iff \omega_{3,4}(p) \neq 0.$$

If $\beta \equiv 0$ along the singular set near p , then

$$\delta(p) \neq 0 \iff \omega_{3,5}(p) \neq 0.$$

Criteria for (3, 4)- and (3, 5)-cuspidal edges

New in this work

Criterion for a (3, 4)-cuspidal edge

Let $f : \Sigma \rightarrow \mathbb{R}^3$ be a frontal and $p \in \Sigma$. Then

$$p \text{ is a (3, 4)-cuspidal edge} \iff \begin{cases} f \text{ is a front near } p, \\ p \text{ is a singular point of the first kind} \\ \text{with degeneracy order 2.} \end{cases}$$

$\cdot$ 3-type edge
 $\cdot \omega_{3,4}(p) \neq 0$

Criterion for a (3, 5)-cuspidal edge

Let p be a singular point of the first kind with degeneracy order 2. Then

$$p \text{ is a (3, 5)-cuspidal edge} \iff \begin{cases} \omega_{3,4} \equiv 0 & \text{on } S(f) \text{ near } p, \\ \omega_{3,5}(p) \neq 0. \end{cases}$$

$\cdot$ 3-type edge
 $\cdot \omega_{3,4} \equiv 0, \omega_{3,5}(p) \neq 0$

Main result III: intrinsic criteria for singularity types

Theorem C

f is a (3,4)-cuspidal edge (72/9/16)

(3-type edge は (変化する))

Let p be a generic singular point of the first kind of degeneracy order 2.
Then

$$f \text{ is a (3,4)-cuspidal edge at } p \iff \beta(p) \neq 0.$$

Moreover,

$$f \text{ is a (3,5)-cuspidal edge at } p \iff \begin{cases} \beta \equiv 0 & \text{on } S(f) \text{ near } p, \\ \delta(p) \neq 0. \end{cases}$$

δ is defined without assuming $\omega_{3,4} \equiv 0$.

As a consequence,

$$\boxed{\text{same } ds^2} \implies \boxed{\text{same intrinsic data } \beta, \delta}$$

$$\implies \boxed{\text{singularity type preserved by isometries}}$$

Main result IV: singularity types are preserved by isometries

As a consequence,

$$\begin{aligned} \boxed{\text{same } ds^2} &\implies \boxed{\text{same intrinsic data } \beta, \delta} \\ &\implies \boxed{\text{singularity type preserved by isometries}} \end{aligned}$$

Corollary D

Let

$$f : \Sigma \rightarrow \mathbb{R}^3, \quad g : \bar{\Sigma} \rightarrow \mathbb{R}^3$$

be locally isometric frontals near corresponding generic singular points p and $\bar{p}$ of the first kind and degeneracy order 2.

Then, for $j = 4, 5$,

$$f \text{ is a } (3, j)\text{-cuspidal edge at } p \iff g \text{ is a } (3, j)\text{-cuspidal edge at } \bar{p}.$$

Summary and outlook

New constructions

- Degeneracy order and NACS for second-order singularities.
- Intrinsic quantities β and δ from the first fundamental form.

Main results

- (i) $\beta \equiv \delta \equiv 0 \iff K$ is bounded.
- (ii) $\beta > 0 \iff$ saddle shape, $\beta < 0 \iff$ convex.
- (iii) β, δ intrinsically characterize generic $(3, 4)$ - and $(3, 5)$ -cuspidal edges.
- (iv) Isometries preserve these two singularity types in the generic class.

Outlook

- Intrinsic classification of non-generic 3-type edges.
- Geometric meaning of δ .

Appendix: Signed limiting normal curvature

Let ξ be a singular vector field and η a null vector field satisfying

$$\eta f = \eta^2 f = \mathbf{0}, \quad \xi f, \eta^3 f \text{ linearly independent on } S(f).$$

New in this work

Definition

For a 3-type edge, define

$$\tilde{\kappa}_\nu(p) := \frac{\det(\xi f, \eta^3 f, \xi^2 f)}{\|\xi f \times \eta^3 f\| \|\xi f\|^2}(p).$$

- $\tilde{\kappa}_\nu$ is independent of the choice of a positively oriented pair (ξ, η) .
- It is the ordinary limiting normal curvature with the normal orientation fixed by (ξ, η) :

$$\tilde{\kappa}_\nu = \pm \kappa_\nu.$$

Appendix: Cuspidal curvatures

Let ξ be a singular vector field and η a null vector field such that

$$\eta f = \eta^2 f = \mathbf{0}, \quad \xi f, \eta^3 f \text{ linearly independent on } S(f).$$

Definition (cuspidal curvatures)

The $(3,4)$ -cuspidal curvature is defined by

$$\omega_{3,4}(p) := \frac{\|\xi f(p)\|^{4/3} \det(\xi f(p), \eta^3 f(p), \eta^4 f(p))}{\|\xi f(p) \times \eta^3 f(p)\|^{7/3}}.$$

The $(3,5)$ -cuspidal curvature is defined by

$$\omega_{3,5}(p) := \frac{\|\xi f(p)\|^{5/3} \det(\xi f(p), \eta^3 f(p), \eta^5 f(p))}{\|\xi f(p) \times \eta^3 f(p)\|^{8/3}}.$$

- $\omega_{3,4}(p)$ is an invariant.
- If $\omega_{3,4} \equiv 0$ along the singular set, then $\omega_{3,5}(p)$ is also an invariant.

Appendix: Product curvatures and β, δ

The cuspidal curvatures $\omega_{3,4}$ and $\omega_{3,5}$ were introduced by Martins–Saji–Santos–Teramoto.

New in this work

Definition (product curvatures in this work)

$$\Pi_{3,4} := \tilde{\kappa}_\nu \omega_{3,4}, \quad \Pi_{3,5} := \tilde{\kappa}_\nu \omega_{3,5}.$$

$\Pi_{3,5}$ is well-defined when $\omega_{3,4} \equiv 0$ along the singular set.

Relation with β, δ

$$\Pi_{3,4} = -\frac{1}{2}\beta,$$

$$\beta \equiv 0 \text{ \& generic } \implies \Pi_{3,5} = -\frac{1}{2}\delta.$$

δ is defined without assuming $\omega_{3,4} \equiv 0$.

Appendix: Transformation formulas between NACSs

Let (u, v) and (ξ, η) be two NACSs near p , and write the coordinate transformation as $u = u(\xi, \eta)$, $v = v(\xi, \eta)$.

Let $\hat{E}$ and $\hat{G}$ denote the coefficients of the first fundamental form with respect to (ξ, η) .

Higher-order transformation formulas between NACSs

Along the singular set $\{v = 0\}$ (equivalently, $\{\eta = 0\}$),

$$\hat{E}_{\eta^3} = v_{\eta} E_{v^3},$$

$$\hat{E}_{\eta^4} = E_{v^4} + 6v_{\eta^2} E_{v^3},$$

$$\hat{E}_{\eta^5} = v_{\eta} E_{v^5} + 10v_{\eta^2} v_{\eta} E_{v^4} + (10v_{\eta^3} + 15\{v_{\eta^2}\}^2 v_{\eta}) E_{v^3},$$

$$\hat{G}_{\eta^5} = v_{\eta} (G_{v^5} + 120v_{\eta^2}),$$

$$\hat{G}_{\eta^6} = G_{v^6} + 27v_{\eta^2} G_{v^5} + 300v_{\eta} v_{\eta^3} + 1170\{v_{\eta^2}\}^2.$$

Here $\hat{E}$, $\hat{G}$, and the derivatives of v are evaluated at $(\xi, 0)$, while E and G are evaluated at $(u(\xi, 0), 0)$.

Appendix: A sign change of β in the generic case

The surface is generic along the whole singular set, but the origin is not a $(3,4)$ -cuspidal edge.

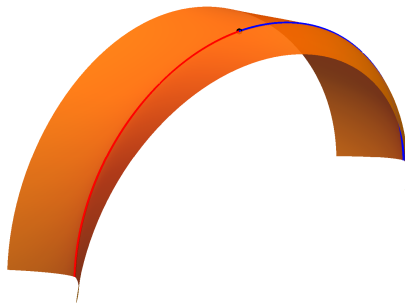


Figure: A surface with $\beta(u) = u$. It has a $(3,4)$ -cuspidal edge for $u \neq 0$.

Appendix: A sign change of β at a non-generic point

The surface has a $(3,4)$ -cuspidal edge at every point of the singular set, but the origin is non-generic.

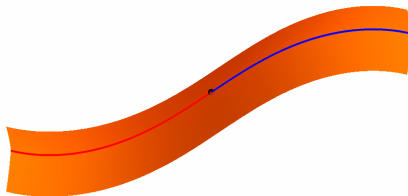


Figure: A $(3,4)$ -cuspidal edge with $\beta(u) = u$. The origin is non-generic.

Appendix: Map formulas for β and δ

Let (u, v) be a local coordinate system satisfying

$$S(f) = \{v = 0\}, \quad f_v(u, 0) = f_{vv}(u, 0) = \mathbf{0}, \quad \text{rank}(f_u, f_{v^3})(u, 0) = 2.$$

For δ , we fix the orientation of ∂_v .

Map formulas for β and δ

The quantities β and δ , originally defined in a NACS, can be written as follows:

$$\beta(u) = -2 \frac{\det(f_u, f_{v^3}, f_{uu}) \det(f_u, f_{v^3}, f_{v^4})}{\|f_u\|^{2/3} \|f_u \times f_{v^3}\|^{10/3}}(u, 0)$$

$$\begin{aligned} \delta(u) = & -2 \frac{\det(f_u, f_{v^3}, f_{uu}) \det(f_u, f_{v^3}, f_{v^5})}{\|f_u\|^{1/3} \|f_u \times f_{v^3}\|^{11/3}}(u, 0) \\ & + \frac{10}{3} \frac{\|f_u\|^{5/3}}{\|f_u \times f_{v^3}\|^{17/3}} \left\{ \langle f_{v^3}, f_{v^4} \rangle - \frac{\langle f_u, f_{v^3} \rangle \langle f_u, f_{v^4} \rangle}{\|f_u\|^2} \right\} \det(f_u, f_{v^3}, f_{uu}) \det(f_u, f_{v^3}, f_{v^4})(u, 0) \\ & + \frac{4}{3} \frac{\|f_u\|^{5/3}}{\|f_u \times f_{v^3}\|^{17/3}} \left\{ \langle f_{uu}, f_{v^3} \rangle - \frac{\langle f_u, f_{uu} \rangle \langle f_u, f_{v^3} \rangle}{\|f_u\|^2} \right\} \{\det(f_u, f_{v^3}, f_{v^4})\}^2(u, 0) \end{aligned}$$

Appendix

Counterexample to Theorem C without the genericity assumption:

$$f(u, v) = \left(u, \frac{u^2}{2} + \frac{v^3}{6}, \frac{v^4}{24} \right)$$

- The singular set is $\{v = 0\}$.
- Along the singular set, $\kappa_\nu = 0$ and $\beta = \mathbf{0}$, but f is a $(3, 4)$ -cuspidal edge.

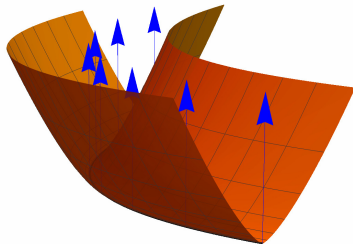


Figure: Counterexample to Theorem C in the non-generic case

Appendix

Singularity type is not necessarily preserved by isometries:



$(4, 5)$ -cuspidal edge



$(4, 5, +7)$ -cuspidal edge

Figure: Isometric surfaces with the same first fundamental form but different singularity types